USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 22–23 LAB - Tapering, Squint, and Quantization
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.7: I can apply amplitude tapering (uniform, cosine, Chebyshev, Taylor) to control sidelobe level and predict the pattern trade-off.

Learning Objective 3.8: I can identify grating lobes, beam squint, and quantization effects in a phased array and describe their impact on system performance.

This packet is the L22–23 Lab, in three parts, and you turn it in once, complete, before Lesson 24. **Part 1, Predict**, is yours to do before the lab period: what each Part 2 table should read on your kit. **Part 2, Measure**, is your own kit's readings, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict.** Work at the HB100 frequency your kit reported with **Find HB100** in an earlier lab. If you did not record one, use the nominal 10.525 GHz and say so. The Pre-Lab frames of Lessons 22 and 23 name the frame that works each method, at 10.0 GHz.

   The array has $N = 8$ patches, E1 to E8, on a $d = 14$ mm pitch. Facing the patches, E1 is on the right and E8 on the left, and $+\theta$ is on the E8 side. The HB100 stands 1.2 m from the array. The Element Gains sliders are voltage amplitudes in percent of full scale, and the presets set them, from the end element inward and mirrored on the other half, to Hann 12, 43, 77, 100; Blackman 6, 27, 66, 100; and Chebyshev 4, 23, 62, 100. Table B is a symmetric taper of your own design, with an HPBW no wider than 16° and the first sidelobe below -20 dBc. Table C leaves Rx1, Rx4, and Rx7 on, then only Rx1 and Rx5. Table D puts the HB100 at 45° and sweeps at a **Signal BW** of 10 and then 500 MHz. Table E holds the beam at 15° with the Blackman taper and sweeps the HB100 in azimuth at 7, 4, 3, and 2 phase bits. Report levels as the tables do: lobe levels in dBc, Table C heights against its 0° lobe, and peak drops against the uniform trace.

   Fill in every **Predicted** cell, the shaded ones, in the setup table and Tables A to E below, and show your work for one row of each table here.

Part 2: Measure (your kit, in class)

Follow the procedure on the L22–23 Lab page and fill in every **Measured** cell, the white ones, in the tables below. Open the **Lab Presets** section before you press a preset. Press **Find HB100**, then **Calibrate**. Set **Rx Gain** so that the uniform main-lobe peak, source at boresight, reads -10 ± 3 dBFS on the FFT, and leave it there. Keep the source 1.2 m from the array, and keep $\pm 40^\circ$ of the kit clear of walls and people. Record each reading as you take it, and every level in dBc: decibels below the main-lobe peak of the same trace, its dBFS reading subtracted, so a level means the same thing on every trace.

Setup	**Predicted**	**Measured**
Signal Freq		
Source distance (far-field distance predicted)		
Rx Gain		
Floor		

2. **Table A: the presets.** Source at boresight, the Tapering preset, Enforce Symmetric Taper on. Freeze the uniform trace as the reference. If the uniform trace's two first sidelobes differ by more than about 2 dB, recalibrate before you go on.

	HPBW		**Peak drop**		**First sidelobe**	
Preset	**Predicted**	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**
Uniform			reference			
Hann						
Blackman (optional)						
Chebyshev (optional)						

3. **Table B: your design.** The Predicted row is the taper you designed in Part 1. Load it, sweep, and adjust until it meets the specification: an HPBW no wider than 16° and the first sidelobe below -20 dBc. The Measured row is your final gains and readings.

	Rx1, Rx8	**Rx2, Rx7**	**Rx3, Rx6**	**Rx4, Rx5**	**HPBW**	**Peak drop**	**First sidelobe**
Predicted							
Measured							

4. **Table C: grating lobes.** Source at boresight, Enforce Symmetric Taper off, the Grating Lobes preset (Rx1, Rx4, Rx7 on). Then Rx4 and Rx7 to 0% and Rx5 to 100%. Read each lobe's angle and its height against the 0° lobe, and the peak drop against the frozen uniform trace.

Elements on	d_{eff}	Lobe angles		Height vs 0°		Peak drop	
		Predicted	Measured	Predicted	Measured	Predicted	Measured
Rx1, Rx4, Rx7	42 mm						
Rx1, Rx5	56 mm						

5. **Table D: beam squint.** Place the HB100 at about 45° on the 1.2 m arc and press the Uniform preset. Set **Signal BW** to 10 MHz, sweep, and record **Angle** (set to **Peak**). Then choose the Beam Squint preset, check that **Signal BW** reads 500 MHz (Calibrate, Find HB100, and the Rect preset all reset it to 0), and sweep again.

Signal BW	Angle (Peak)		Shift	
	Predicted	Measured	Predicted	Measured
10 MHz			reference	
500 MHz				

6. **Table E: phase quantization.** The Quantization preset holds the beam at 15° with the Blackman taper and opens Signal vs Time. Sweep the HB100 in azimuth from -90° to $+90^\circ$ on the 1.2 m arc, in about 20 s, at Phase Shift Bits 7, 4, 3, and 2, with Use Bits checked before each sweep. On each trace read the strongest lobe outside the main lobe, in dBc, and where it is. Write "floor" where nothing rises out of it.

Bits	Strongest lobe outside the main lobe		Where	
	Predicted	Measured	Predicted	Measured
7				
4				
3				
2				

Before you leave: Use Bits on, Phase Shift Bits 7, Signal BW 10 MHz, and the Uniform preset.

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

7. **The tapers.** Compare your taper measurements (Tables A and B) with your predictions. Build an error budget that corresponds with your measurements and explains any disagreement. Then say which taper you would use on this kit and why, and what your peak-drop readings don't explain.

8. **Grating lobes and squint.** Compare your grating-lobe measurement (Table C) and your beam-squint measurement (Table D) with your predictions. Explain each difference your measurement can resolve, and say what you would change in either measurement to test your explanation.

9. **Phase bits.** Compare your four quantization sweeps (Table E) with each other and with your predictions. From your data, state when rounding the phases matters on this array and when it does not, and what your four sweeps don't explain.

Documentation: _____